

DATA ANALYTICS PROJECT

E-Commerce Data Analytics Project

SQL

Excel

Power BI

Prepared By

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Aspiring Data Analyst

Project Overview

DATASET DETAILS

DATASET	Global Superstore
PERIOD	2011 – 2014
RECORDS	~51,271 Rows
MARKETS	7 Global Markets
CATEGORIES	3 Product Categories

TOOLS USED

SQL

Excel

Power BI

Project Objectives



Analyze Sales Performance across 4 years and 7 markets



Evaluate Profitability by category, segment & sub-category



Understand Customer Behavior and buying patterns



Identify Business Growth Opportunities across markets



Create Interactive Dashboards in Excel & Power BI

Data Cleaning & Preparation



Duplicate Removal

Identified and removed duplicate entries to ensure data integrity



Missing Values

Checked and handled all null/missing fields appropriately



Format Standardization

Normalized date formats, column names, and data types



Data Quality Check

Verified accuracy and consistency across all 51,271 records



Dataset Preparation

Exported clean dataset for SQL, Excel & Power BI analysis

CLEANED DATASET STATS

51,271

Total Records

24

Columns

2011–2014

Time Period

7

Markets

SQL Analysis Overview

10 Key Business Questions Analyzed Using SQL

1

Total Sales Analysis

2

Total Profit Analysis

3

Sales by Region / Market

4

Profit by Product Category

5

Top Customers by Sales

6

Top Products by Sales

7

Orders with Negative Profit

8

Monthly Sales Trend

9

Shipping Mode Analysis

10

Market Revenue Analysis



All queries executed on MySQL using the cleaned Global Superstore dataset imported via Python scripting.

Key Insights from SQL Analysis



APAC market generated the highest revenue across all years



Technology category produced the highest profit margin



Standard Class was the most frequently used shipping mode (60%)



Consumer segment contributed the largest portion of total sales



Top customers generated a significant share of total revenue

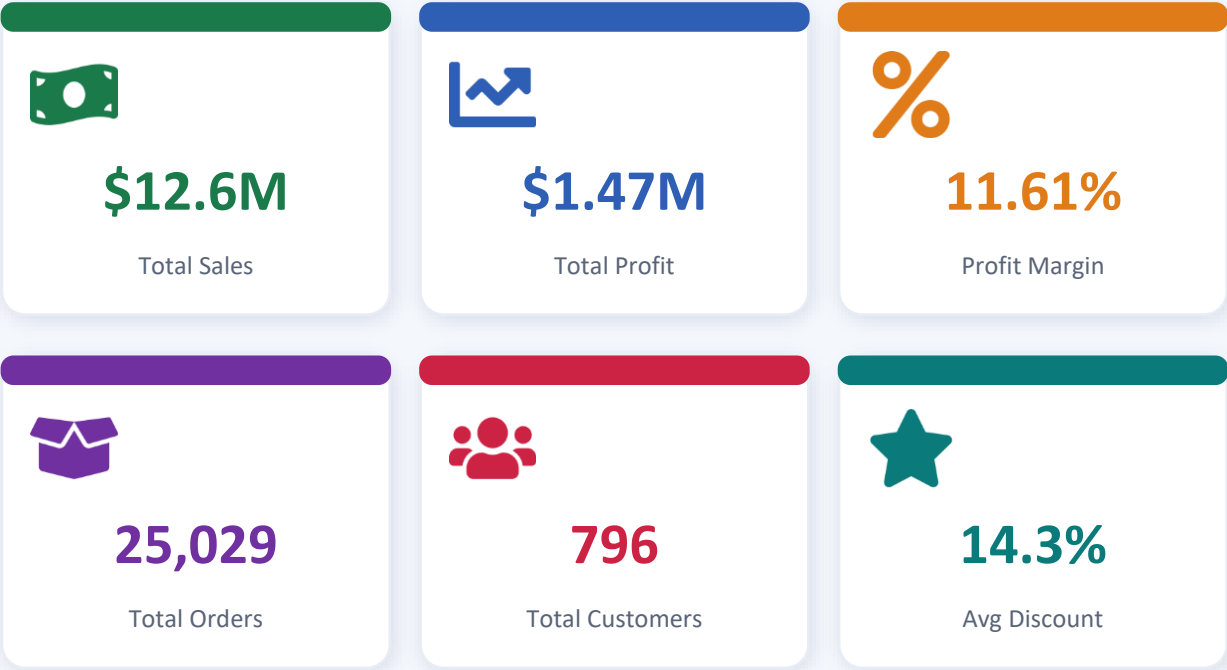


Sales showed a strong upward trend throughout 2011–2014



Several transactions recorded negative profit, flagging discounting issues

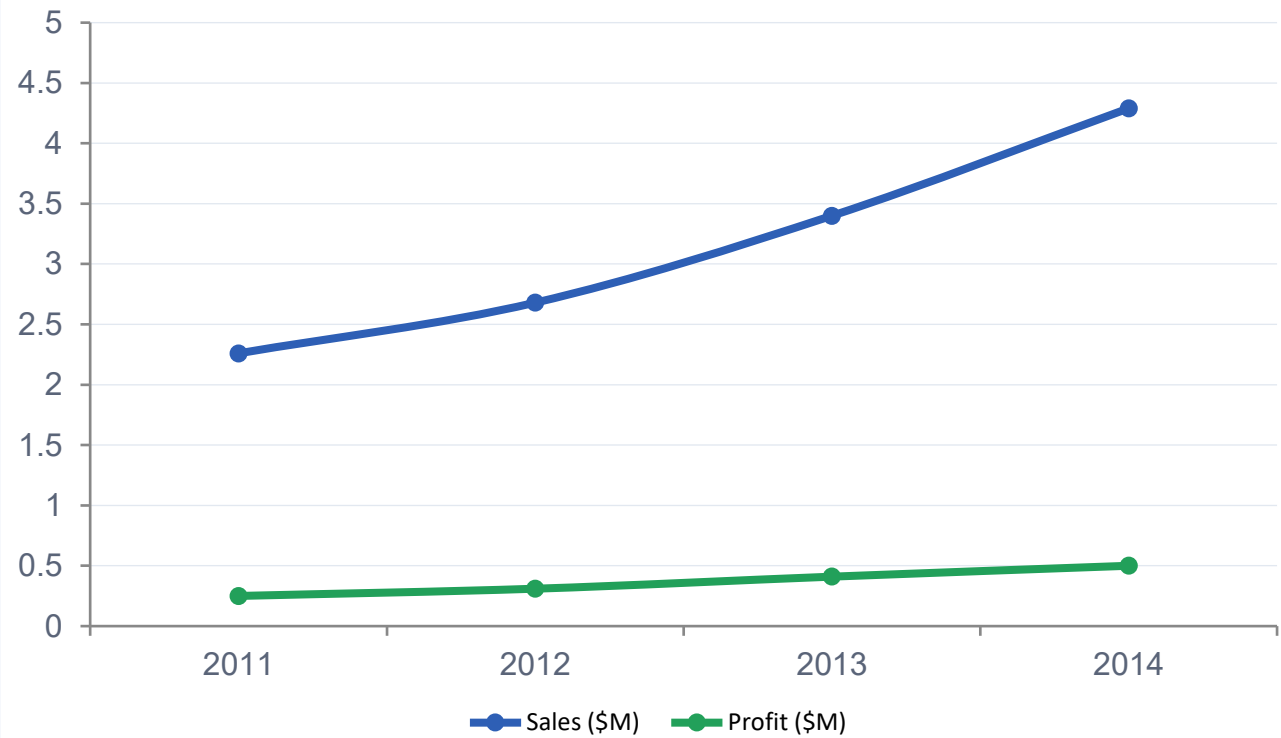
Excel KPI Dashboard



Dashboard Components	
✓	Year-Over-Year Performance Table
✓	Sales & Profit by Market
✓	Sales by Product Category
✓	Sales by Customer Segment
✓	Shipping Mode Analysis
✓	Top 5 Sub-Categories by Profit
✓	Bottom 5 Sub-Categories by Loss

Business Performance Insights

Sales & Profit Trend (2011–2014)



90% Sales Growth over 4 years (\$2.3M → \$4.3M)



Consistent profit improvement year on year



796 unique customers maintained throughout the period



Avg discount of 14.3% influenced overall margins



Revenue growth supported by 7 global markets

Excel Pivot Analysis



Monthly Sales Trend

Line chart showing sales pattern month by month



Sales by Region / Market

Performance across 7 global markets



Profit by Category

Technology, Furniture & Office Supplies



Segment-wise Sales

Consumer, Corporate & Home Office segments



Top Customers

Highest revenue-generating customers



Shipping Mode Analysis

Order distribution across ship modes



Top 5 Profitable Sub-Categories

Copiers, Phones, Bookcases lead profits



Bottom 5 Sub-Categories

Tables shows highest loss (-\$64K)

Power BI Interactive Dashboard



Sales Trend Analysis – Year & Month



Regional Market Performance Map



Customer Segment Breakdown



Category & Sub-Category Performance



Profit Analysis with Drill-down



KPI Cards – Sales, Profit, Margin



Dynamic Slicers & Cross-Filters

Business Recommendations



Focus on High-Profit Categories

Prioritize Technology products which yield highest profit margins.



Fix Loss-Making Sub-Categories

Audit Tables sub-category which has -\$64K negative profit.



Expand Top-Performing Markets

Double down on APAC & EU markets generating 50%+ of revenue.



Strengthen Customer Retention

Launch loyalty programs for high-value Consumer segment customers.



Optimize Discount Strategy

Review discount policies causing negative-margin transactions.



Operational Efficiency

Reduce over-reliance on Standard Class; explore faster options.

Project Summary

Achievements



SQL Data Analysis Completed

10 business queries executed



Excel KPI Dashboard Developed

6 KPIs + 7 analysis tables



Pivot Analysis Completed

7 pivot analyses performed



Power BI Dashboard Developed

3 pages with 8+ visuals



Business Insights Generated

Data-driven findings documented



Strategic Recommendations Prepared

6 actionable recommendations

CONCLUSION



This project successfully transformed raw Global Superstore data into actionable business intelligence.



\$12.6M Sales Analyzed



7 Markets Covered



4-Year Trends Identified



6 Recommendations Delivered

Thank You!

Thank you for providing me this opportunity to complete this project.

I am passionate about Data Analytics and my goal is to build a successful career as a Data Analyst in a reputed organization. If there are any suitable opportunities, I would be grateful if my profile could be considered.

KUMARI DUVVARAPU

Aspiring Data Analyst • SQL | Excel | Power BI

Data Cleaning

SQL Analysis

Excel Dashboards

Power BI

Data Visualization